CADI · MILESTONE

Hierarchical Reasoning & Graded Recognition

Recognition by reasoning — transparent, auditable, and honest about uncertainty.

CADI · INTELAVATION LLC AND WESAVE

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THE RESULT

CADY reasons within a structured hierarchy of knowledge — and knows the limits of her own certainty. Given an object's properties, she climbs her own taxonomy to assemble what it is, classifies it, and shows her work; and when the evidence only reaches a general category, she says so honestly ("it is a vehicle, but I cannot say which kind") instead of guessing.

HOW IT WAS PROVEN

CADY was taught a two-branch, three-level taxonomy of everyday objects — vehicles and tableware — with shared traits at the parent level and distinguishing traits at each item. She was then given objects described *only by their properties*, never told what they were, and asked to reason out their identity. The demonstration ran live over her saved knowledge graph; her FPGA reasoning core was untouched throughout — her learned-pattern count held constant at every step, a verifiable sign the demonstration rode her knowledge layer without disturbing her cognition core.

WHAT WAS PROVEN — FIVE TESTS, ALL PASSED

1. **Climbs the hierarchy to know an object's full nature.** Told an item has wheels, four-plus wheels, and an enclosed cabin, she concludes *car* — showing that "wheels" is *inherited* from the broader category *vehicle*, while "enclosed cabin" is the car's own distinguishing trait. She reasoned upward through the taxonomy, not off a flat list.
2. **Discriminates on the right trait among near-identical siblings.** A motorcycle and a bicycle share almost everything. She separates them on the *one* trait that differs — motor vs. pedal-powered — and states that distinction explicitly.
3. **Supplies missing knowledge by reasoning.** Told only "utensil, elongated, bowl-end," she concludes *spoon* — supplying a fact she was never given ("used for eating") by recognizing the traits belong to tableware and inheriting the category's properties.
4. **Refuses to guess when the evidence is insufficient.** Given traits that fit *either* a motorcycle or a bicycle but without the deciding one, she does not pick. She reports it is consistent with both and names the exact missing trait that would decide it.
5. **Places things at the right level of certainty — the headline result.** Told only that something "has wheels," she concludes: it is a vehicle, but the specific kind cannot be determined — and names exactly what she would need to narrow it down.

WHAT IT MEANS

1. **Recognition by reasoning, not memorization.** A neural net abstracts objects into opaque weights nobody can inspect. CADY reaches the same kind of conclusion by reasoning over named, readable properties — every step auditable. Ask her *why* she concluded "car" and you get a real answer that traces to specific facts. The capability a black box hides, made transparent.
2. **Graded, honest confidence — a differentiator and a safety property.** A conventional AI forced to classify outputs "motorcycle, 41%" and buries the fact that it is really only sure the thing is *a vehicle*. CADY tells you precisely how far her certainty reaches. A system that knows the limit of its own evidence — and says so — is exactly what high-stakes domains (finance, security, infrastructure, defense) require. It does not bluff.
3. **It mirrors how reasoning actually works.** People recognize hierarchically: wheels → vehicle → car → sedan as detail arrives. CADY demonstrably does the first stages of this — in plain logic you can read. Structured cognition, not pattern lookup.
4. **It builds on everything proven before it.** This is the fourth proven rung in a coherent progression, each step adding capability on the same untouched reasoning core. Not a one-off demo — a system getting provably more capable along a deliberate path.

THE PROVEN LADDER

Each rung was proven live, on the same reasoning core, each auditable and built additively on the last.

1. **The cognition loop.** She learns a fact, retains it after the source is removed, refines it with new information, and reasons across her memories to a conclusion no one handed her.
2. **Reasoning to a verdict.** She applies a learned rule to classify something by its properties, explains which rule fired, and refuses to guess when the evidence is insufficient.
3. **Learning from tools, at scale.** A real perception tool labeled 60 distinct objects; she learned all 60 as her own knowledge; the tool was removed; she retained and recalled all 60.
4. **Hierarchical reasoning & graded recognition.** This milestone — she reasons within a multi-level taxonomy, assembles an object's full nature from inherited and specific traits, and honestly places things at the right level of certainty.

The shape of the story: from "learns a fact" to "reasons within a structured hierarchy of knowledge, and knows what it does not know" — every step transparent, every step on owned silicon, every step proven.

THE BOUNDARY — PRESERVED

What this milestone is: transparent, auditable reasoning over properties — from an object's traits to its identity, or honestly to its category, within a taught hierarchy, showing every step.

What it is not: it is not "she sees an object and identifies it on her own." The properties were provided. Perceiving them directly from raw vision — looking at a spoon and extracting "elongated, bowl-end" herself — is the genuine frontier and the next major build. Nor is it a neural-net-style learned abstraction: the reasoning is over named facts, by design.

CADY reasons transparently over an object's properties to identify it and to know the limits of her certainty; perceiving those properties directly from raw vision is the next frontier, and the focus of the funded roadmap.

Proven 2026-06-30, live on CADY's reasoning system; standalone read-only verification over her knowledge graph; reasoning core untouched. The full technical detail — taxonomy, rules, classifier logic, and exact test outputs — is recorded separately and available as a verifiable appendix.

LEADERSHIP

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